**Topic** - Introduction to Operating system

**Objective –** To understand the structure and organization of the file system

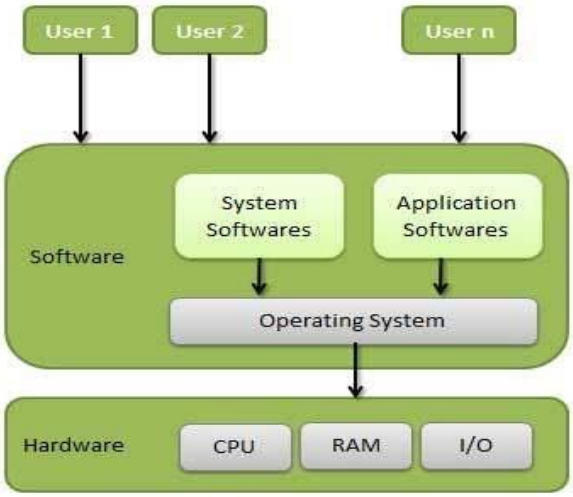
**Outcome -** Students will learn how Operating System is Important for Computer System

## INTRODUCTION TO OPERATING SYSTEM

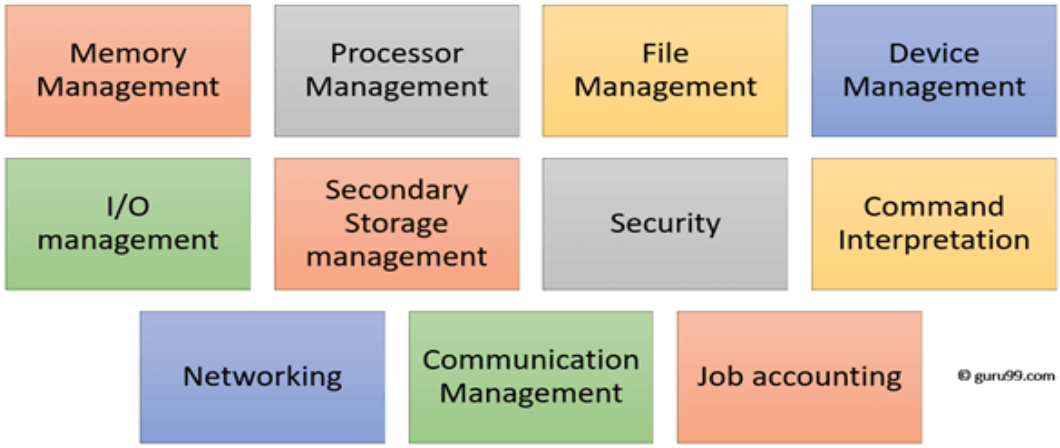
An operating system (OS) is the program that, after being initially loaded into the computer by a boot program, manages all of the other application programs in a computer.

An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.

**Architecture**




## Functions of Operating System



1. **Process management**: Process management helps OS to create and delete processes. It also provides mechanisms for synchronization and communication among processes.

2. **Memory management:** Memory management module performs the task of allocation and de-allocation of memory space to programs in need of this resources.

3. **File management**: It manages all the file-related activities such as organization storage, retrieval, naming, sharing, and protection of files.

4. **Device Management**: Device management keeps tracks of all devices. This module also responsible for this task is known as the I/O controller. It also performs the task of allocation and de-allocation of the devices.

5. **I/O System Management:** One of the main objects of any OS is to hide the peculiarities of that hardware devices from the user.

6. **Secondary-Storage Management**: Systems have several levels of storage which includes primary storage, secondary storage, and cache storage. Instructions and data must be stored in primary storage or cache so that a running program can reference it.

7. **Security**: Security module protects the data and information of a computer system against malware threat and authorized access.

## Features of Operating System (OS)

Here is a list important features of OS:

- Protected and supervisor mode
- Allows disk access and file systems Device drivers Networking Security
- Program Execution
- Memory management Virtual Memory Multitasking
- Handling I/O operations
- Manipulation of the file system
- Error Detection and handling
- Resource allocation
- Information and Resource Protection

## Components of Operating System

## USER MODE VS. KERNEL MODE

### User Mode

The system is in user mode when the operating system is running a user application such as handling a text editor. The transition from user mode to kernel mode occurs when the application requests the help of operating system or an interrupt or a system call occurs.

The mode bit is set to 1 in the user mode. It is changed from 1 to 0 when switching from user mode to kernel mode.

### Kernel Mode

The system starts in kernel mode when it boots and after the operating system is loaded, it executes applications in user mode. There are some privileged instructions that can only be executed in kernel mode.

The mode bit is set to 0 in the kernel mode. It is changed from 0 to 1 when switching from kernel mode to user mode.

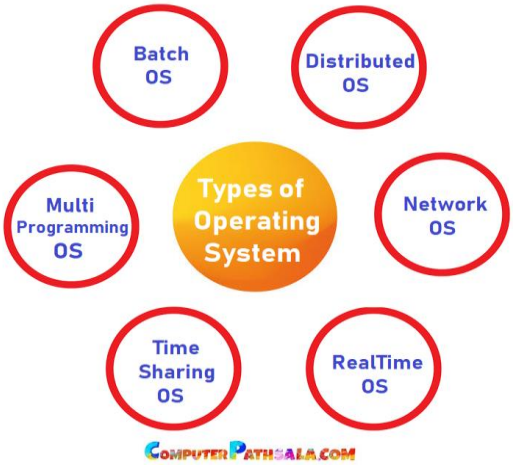
**Summary -** An Operating System is an interface that helps a user connect and communicate with computer hardware. It is an essential part of computing devices like mobiles, laptops, cameras, and other appliances. Even some washing machines and refrigerators have them.

**Topic** - Types of Operating system

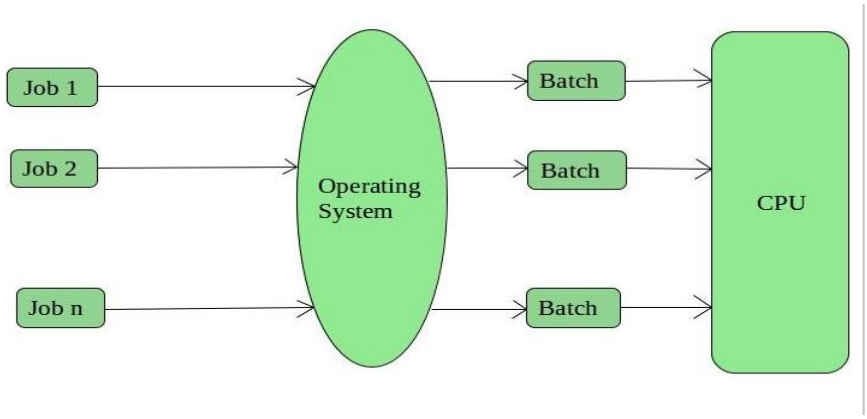
**Objective** – To understand various types of operating system

**Outcome** - Students will learn how processes get execute in different ways in operating system

TYPES :




**1.) BATCH OPERATING SYSTEM-** This type of operating system does not interact with the computer directly. There is an operator which takes similar jobs having the same requirement and group them into batches. It is the responsibility of the operator to sort jobs with similar needs.




**Advantages of Batch Operating System:**

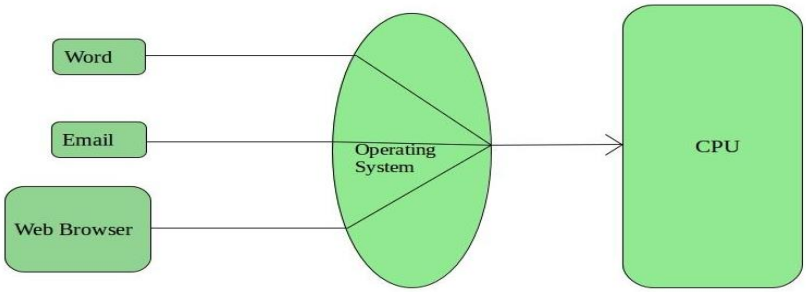
- It is very difficult to guess or know the time required for any job to complete. Processors of the batch systems know how long the job would be when it is in queue
- Multiple users can share the batch systems
- The idle time for the batch system is very less
- It is easy to manage large work repeatedly in batch systems

**Disadvantages of Batch Operating System:**

- The computer operators should be well known with batch systems
- Batch systems are hard to debug

- It is sometimes costly
- The other jobs will have to wait for an unknown time if any job fails

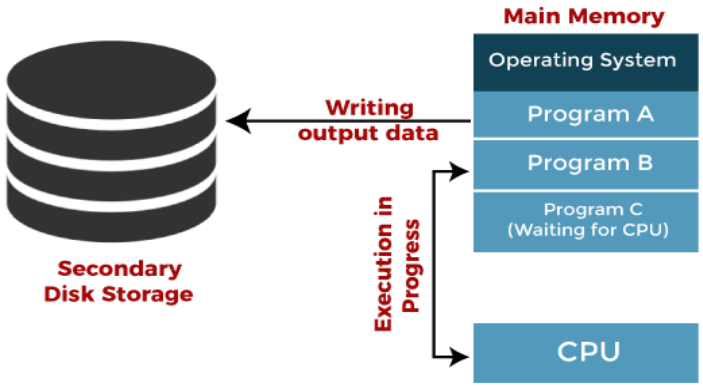
2.) **TIME-SHARING OPERATING SYSTEMS –**
Each task is given some time to execute so that all the tasks work smoothly. Each user gets the time of CPU as they use a single system. These systems are also known as Multitasking Systems. The task can be from a single user or different users also.




**Advantages of Time-Sharing OS:**
- Each task gets an equal opportunity
- Fewer chances of duplication of software
- CPU idle time can be reduced

**Disadvantages of Time-Sharing OS:**
- Reliability problem
- One must have to take care of the security and integrity of user programs and data
- Data communication problem

**3.) MULTIPROGRAMMING OPERATING SYSTEM**
In a multiprogramming environment, when a process does its I/O, The CPU can start the execution of other processes. Therefore, multiprogramming improves the efficiency of the system.




*Jobs in multiprogramming system*

Advantages of Multiprogramming OS

- Throughout the system, it increased as the CPU always had one program to execute.

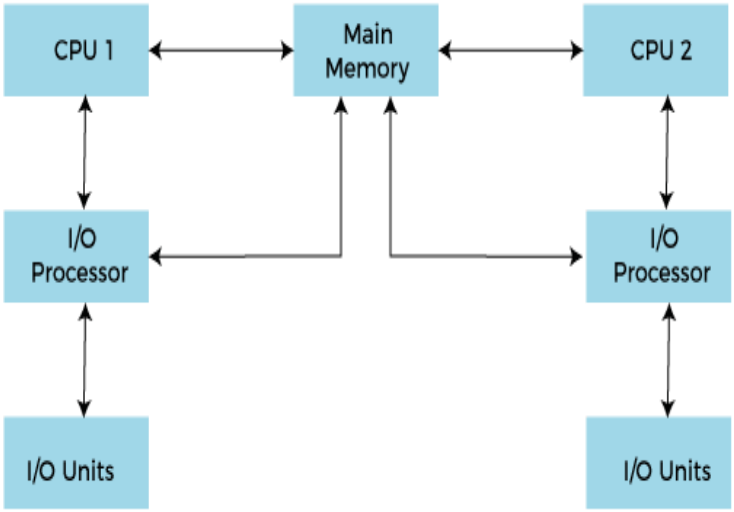
- Response time can also be reduced.

**Disadvantages of Multiprogramming OS**

- Multiprogramming systems provide an environment in which various systems resources are used efficiently, but they do not provide any user interaction with the computer system.

**4.) MULTIPROCESSING OPERATING SYSTEM**

In Multiprocessing, Parallel computing is achieved. There are more than one processors present in the system which can execute more than one process at the same time. This will increase the throughput of the system.




Working of Multiprocessor System

**Advantages of Multiprocessing operating system:**

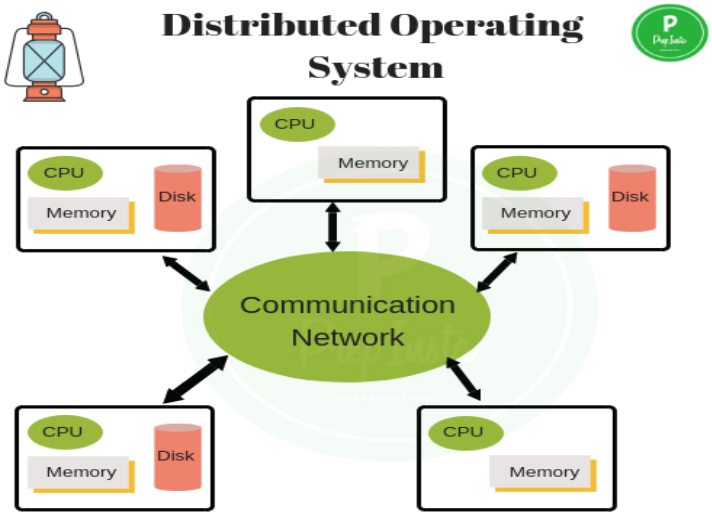
- **Increased reliability:** Due to the multiprocessing system, processing tasks can be distributed among several processors. This increases reliability as if one processor fails, the task can be given to another processor for completion.
- **Increased throughout:** As several processors increase, more work can be done in less.

**Disadvantages of Multiprocessing operating System**

- Multiprocessing operating system is more complex and sophisticated as it takes care of multiple CPUs simultaneously.

**5.) DISTRIBUTED OPERATING SYSTEM**

Distributed systems use multiple central processors to serve multiple real-time applications and multiple users. Data processing jobs are distributed among the processors accordingly.




**The advantages of distributed systems are as follows −**

- With resource sharing facility, a user at one site may be able to use the resources available at another.
- Speedup the exchange of data with one another via electronic mail.
- If one site fails in a distributed system, the remaining sites can potentially continue operating.
- Better service to the customers.
- Reduction of the load on the host computer.
- Reduction of delays in data processing.

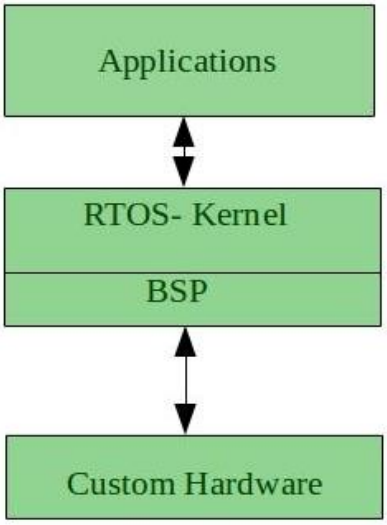
## 6.) REAL-TIME OPERATING SYSTEM –

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**.
**Real-time systems** are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

**Two types of Real-Time Operating System which are as follows:**

- **Hard Real-Time Systems:**
  These OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or airbags which are required to be readily available in case of any accident. Virtual memory is rarely found in these systems.

- **Soft Real-Time Systems:**
  These OSs are for applications where for time-constraint is less strict.




**Advantages of RTOS:**

- **Maximum Consumption:** Maximum utilization of devices and system, thus more output from all the resources
- **Task Shifting:** The time assigned for shifting tasks in these systems are very less.
- **Error free :** These types of systems are error-free.

## Disadvantages of RTOS:

- **Limited Tasks:** Very few tasks run at the same time and their concentration is very less on few applications to avoid errors.
- **Use heavy system resources:** Sometimes the system resources are not so good and they are expensive as well.
- **Thread priority :** It is not good to set thread priority as these systems are very less prone to switching tasks.

**Summary -** The operating system is used to run different software on a computer. All operating systems have one common goal- **to manage and organize the resources and processes of a computer**. These resource managers include processes, threads, files, devices, and networks.

**Topic** - Process control block

**Objective –** It helps you to store all the information required to keep track of all the running processes

**Outcome -** Students will learn how Operating System can easily distinguish between processes.

**DIFFERENCE BETWEEN PROCESS AND PROGRAM**

| Features | Process | Program |
|---|---|---|
| **Definition** | A program has a collection of instructions designed to accomplish a certain task. | A process is an example of an execution program. |
| **Nature** | It is an active entity. | It is a passive entity. |
| **Lifespan** | It has a limited lifespan. | It has a much higher lifespan. |
| **Resources** | It has a high resources requirement, and it requires including CPU, memory address, disk, Input/output during its lifetime. | It doesn't have any resources requirements; it only needs memory space to store the instructions. |
| **Creation** | The new process needs duplication of the parent process. | No much duplication is needed. |
| **Computation Time** | The process takes a long time to access and compute a single fact. | It has no computation time and cost. |
| **Required Process** | It holds resources including CPU, disk, memory address, Input/Output etc. | The program is stored on a disk in a file and doesn't need any additional resources. |
| **Overhead** | It has considerable overhead. | It has no significant overhead cost. |
| **Cache Data** | It may use the cache to store the retrieve the data as it uses OS paging scheme and cache replacement policy like FCFS, LRU, RR, LIFO. | It has the instruction to use cache for its data. |

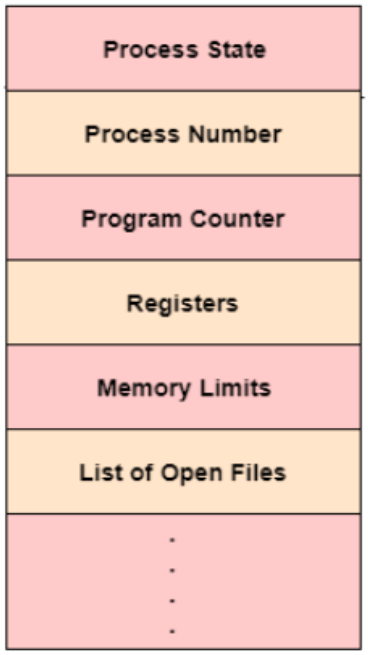
## PROCESS CONTROL BLOCK

Process Control Block is a data structure that contains information of the process related to it. The process control block is also known as a task control block, entry of the process table, etc.

It is very important for process management as the data structuring for processes is done in terms of the PCB. It also defines the current state of the operating system.

### Structure Of The Process Control Block

The process control stores many data items that are needed for efficient process management.




Process Control Block (PCB)

The following are the data items −

**Process State**

This specifies the process state i.e. new, ready, running, waiting or terminated.

**Process Number**

This shows the number of the particular process.

**Program Counter**

This contains the address of the next instruction that needs to be executed in the process.

**Registers**

This specifies the registers that are used by the process. They may include accumulators, index registers, stack pointers, general purpose registers etc.

**List of Open Files**

These are the different files that are associated with the process

**CPU Scheduling Information**

The process priority, pointers to scheduling queues etc. is the CPU scheduling information that is contained in the PCB. This may also include any other scheduling parameters.

**Memory Management Information**

The memory management information includes the page tables or the segment tables depending on the memory system used. It also contains the value of the base registers, limit registers etc.

**I/O Status Information**

This information includes the list of I/O devices used by the process, the list of files etc.

**Accounting information**

The time limits, account numbers, amount of CPU used, process numbers etc. are all a part of the PCB accounting information.

**Location of the Process Control Block**

The process control block is kept in a memory area that is protected from the normal user access. This is done because it contains important process information. Some of the operating systems place the PCB at the beginning of the kernel stack for the process as it is a safe location.

**Summary -** Process Control Block (PCB) is a data structure maintained by the operating system to store information of each process. Role of PCB is to give identity to each process so that Operating System can easily distinguish between processes. The process control block is kept in a memory area that is protected from the normal user access. This is done because it contains important process information.

**Topic** - CPU Scheduling algorithm

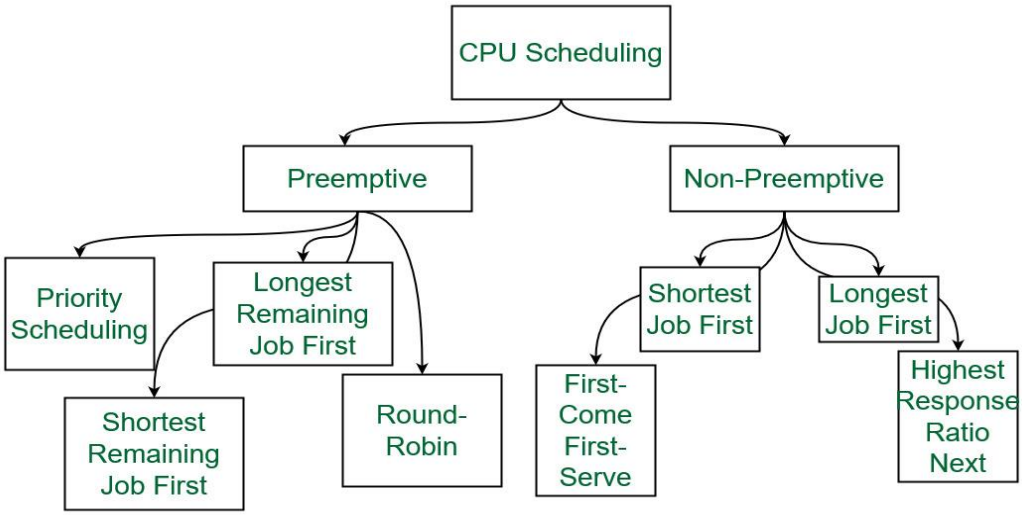
**Objective –** Its main objective is to increase system performance in accordance with the chosen set of criteria.

**Outcome -** Students will learn different scheduling algorithm to run processes.

**WHAT IS CPU SCHEDULING?**

CPU Scheduling is a process of determining which process will own CPU for execution while another process is on hold. The main task of CPU scheduling is to make sure that whenever the CPU remains idle, the OS at least select one of the processes available in the ready queue for execution. The selection process will be carried out by the CPU scheduler. It selects one of the processes in memory that are ready for execution.

**TYPES OF CPU SCHEDULING**




## When scheduling is Preemptive or Non-Preemptive?

To determine if scheduling is preemptive or non-preemptive, consider these four parameters:

1. A process switches from the running to the waiting state.
2. Specific process switches from the running state to the ready state.
3. Specific process switches from the waiting state to the ready state.
4. Process finished its execution and terminated.

## Difference between Preemptive scheduling and Non-Preemptive scheduling

| Parameter | PREEMPTIVE SCHEDULING | NON-PREEMPTIVE SCHEDULING |
|---|---|---|
| Basic | In this resources(CPU Cycle) are allocated to a process for a limited time. | Once resources(CPU Cycle) are allocated to a process, the process holds it till it completes its burst time or switches to waiting state. |
| Interrupt | Process can be interrupted in between. | Process can not be interrupted until it terminates itself or its time is up. |
| Starvation | If a process having high priority frequently arrives in the ready queue, a low priority process may starve. | If a process with a long burst time is running CPU, then later coming process with less CPU burst time may starve. |
| Overhead | It has overheads of scheduling the processes. | It does not have overheads. |
| Flexibility | flexible | rigid |
| Cost | cost associated | no cost associated |
| CPU Utilization | In preemptive scheduling, CPU utilization is high. | It is low in non preemptive scheduling. |
| Waiting Time | Preemptive scheduling waiting time is less. | Non-preemptive scheduling waiting time is high. |
| Response Time | Preemptive scheduling response time is less. | Non-preemptive scheduling response time is high. |
| Examples | Examples of preemptive scheduling are Round Robin and Shortest Remaining Time First. | Examples of non-preemptive scheduling are First Come First Serve and Shortest Job First. |

## CPU scheduling terminologies/Criteria

- **Burst Time/Execution Time:** It is a time required by the process to complete execution. It is also called running time.
- **Arrival Time:** when a process enters in a ready state
- **Finish Time:** when process complete and exit from a system

- **Waiting time:** Waiting time is an amount that specific process needs to wait in the ready queue.
  **Formula: {Turn around time – Burst time}**

- **Response time:** It is an amount to time in which the request was submitted until the first response is produced.
  **Formula: {The time at which a process get CPU first time – Arrival time}**

- **Turnaround Time:** Turnaround time is an amount of time to execute a specific process. It is the calculation of the total time spent waiting to get into the memory, waiting in the queue and, executing on the CPU. The period between the time of process submission to the completion time is the turnaround time.
  **Formula: { Completion time – Arrival time}**

**Summary -** Process Control Block (PCB) is a data structure maintained by the operating system to store information of each process. Role of PCB is to give identity to each process so that Operating System can easily distinguish between processes. The process control block is kept in a memory area that is protected from the normal user access. This is done because it contains important process information.

**Topic -** Non-Preemptive Scheduling algorithm (FCFS, SJF, LJF, HRRN, Multilevel queue)

**Objective –** Its main objective is to increase system performance in accordance with the chosen set of criteria.

**Outcome -** Students will learn to apply different scheduling algorithm to run processes.

## 1.) FCFS SCHEDULING (FIRST COME FIRST SERVED)

In FCFS Scheduling,

- The process which arrives first in the ready queue is firstly assigned the CPU.
- In case of a tie, process with smaller process id is executed first.
- It is always non-preemptive in nature.

**Advantages-**

- It is simple and easy to understand.
- It can be easily implemented using queue data structure.
- It does not lead to starvation.

**Disadvantages-**

- It does not consider the priority or burst time of the processes.
- It suffers from **convoy effect**.

**<u>Convoy Effect</u>**

In convoy effect,

- Consider processes with higher burst time arrived before the processes with smaller burst time.
- Then, smaller processes have to wait for a long time for longer processes to release the CPU.

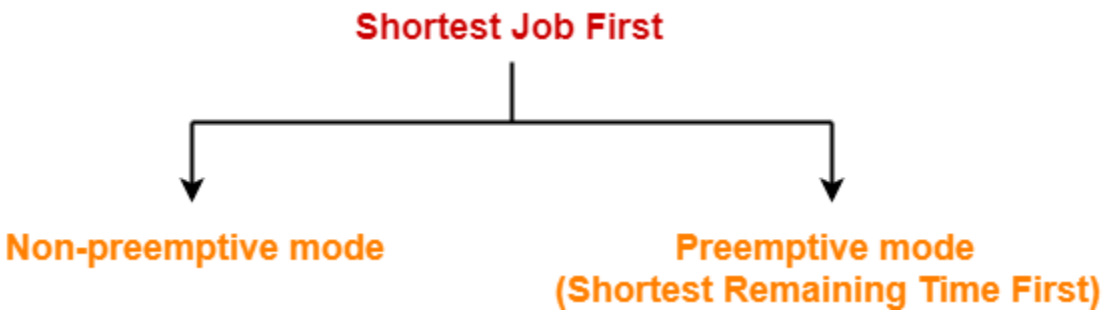
**Example:**

Consider the set of 5 processes whose arrival time and burst time are given below-

## SJF (SHORTEST JOB FIRST) SCHEDULING

In SJF Scheduling,

- Out of all the available processes, CPU is assigned to the process having smallest burst time.
- In case of a tie, it is broken by **FCFS Scheduling**.




- SJF Scheduling can be used in both preemptive and non-preemptive mode.
- Preemptive mode of Shortest Job First is called as **Shortest Remaining Time First (SRTF)**.

**Advantages-**

- SRTF is optimal and guarantees the minimum average waiting time.
- It provides a standard for other algorithms since no other algorithm performs better than it.

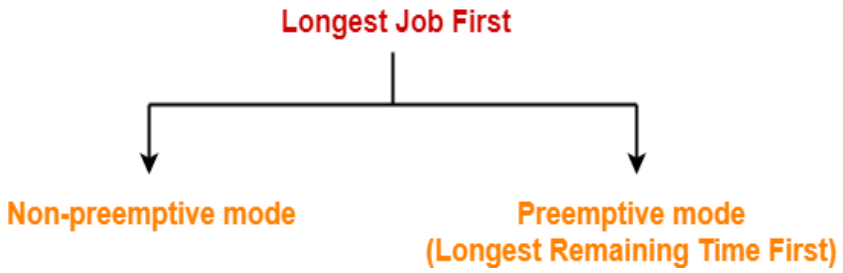
**Disadvantages-**

- It can not be implemented practically since burst time of the processes can not be known in advance.
- It leads to starvation for processes with larger burst time.
- Priorities can not be set for the processes.
- Processes with larger burst time have poor response time.

## LRF (LONGEST JOB FIRST) SCHEDULING ALGORITHM

In LJF Scheduling,

- Out of all the available processes, CPU is assigned to the process having largest burst time.
- In case of a tie, it is broken by FCFS Scheduling.




- LJF Scheduling can be used in both preemptive and non-preemptive mode.
- Preemptive mode of Longest Job First is called as Longest Remaining Time First (LRTF).

**Advantages-**

- No process can complete until the longest job also reaches its completion.
- All the processes approximately finishes at the same time.

**Disadvantages-**

- The waiting time is high.
- Processes with smaller burst time may starve for CPU.

## HRRN (Highest Response Ratio Next) Scheduling Algorithm

In HRRN Scheduling,

- Out of all the available processes, CPU is assigned to the process having highest response ratio.
- In case of a tie, it is broken by **FCFS Scheduling**.
- It operates only in non-preemptive mode.

### Calculating Response Ratio-

Response Ratio (RR) for any process is calculated by using the formula-

**Advantages-**

- It performs better than **SJF Scheduling**.
- It not only favors the shorter jobs but also limits the waiting time of longer jobs.

**Disadvantages-**

- It can not be implemented practically.
- This is because burst time of the processes can not be known in advance.

## MULTILEVEL QUEUE

It may happen that processes in the ready queue can be divided into different classes where each class has its own scheduling needs.

In Multilevel Feedback Queue Scheduling (MFQS), the processes can move between the queues. There are multiple CPUs available for load sharing. This process helps avoid starvation as if there is a process waiting in the low priority queue can be moved to the high priority queue, and a process taking too much of the CPU time will be shifted to the lower priority queue.

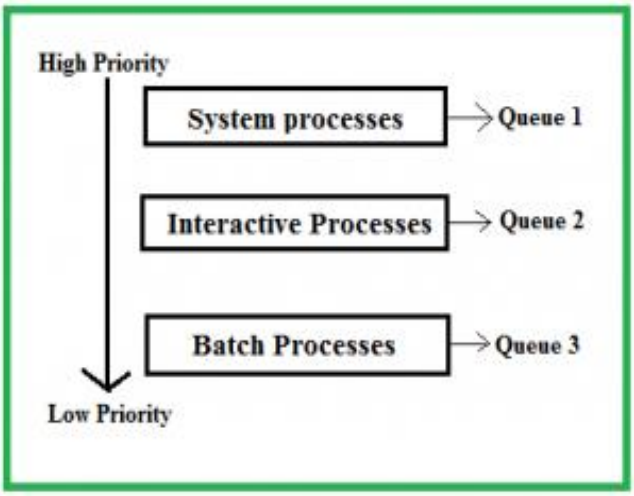
**Advantages of Multilevel Queue CPU Scheduling:**

- The processes are permanently assigned to the queue, so it has advantage of low scheduling overhead.

**Disadvantages of Multilevel Queue CPU Scheduling:**

- Some processes may starve for CPU if some higher priority queues are never becoming empty.
- It is inflexible in nature.

**Ready Queue** is divided into separate queues for each class of processes. For example, let us take three different types of processes System processes, Interactive processes, and Batch Processes. All three processes have their own queue.




The Description of the processes in the above diagram is as follows:

- **System Processes:** The CPU itself has its own process to run which is generally termed as System Process.
- **Interactive Processes**: An Interactive Process is a type of process in which there should be same type of interaction.
- **Batch Processes:** Batch processing is generally a technique in the Operating system that collects the programs and data together in the form of the batch before the processing starts.

**Summary -** Non-preemptive Scheduling is used when a process terminates, or a process switches from running to the waiting state. In this scheduling, once the resources (CPU cycles) are allocated to a process, the process holds the CPU till it gets terminated or reaches a waiting state.